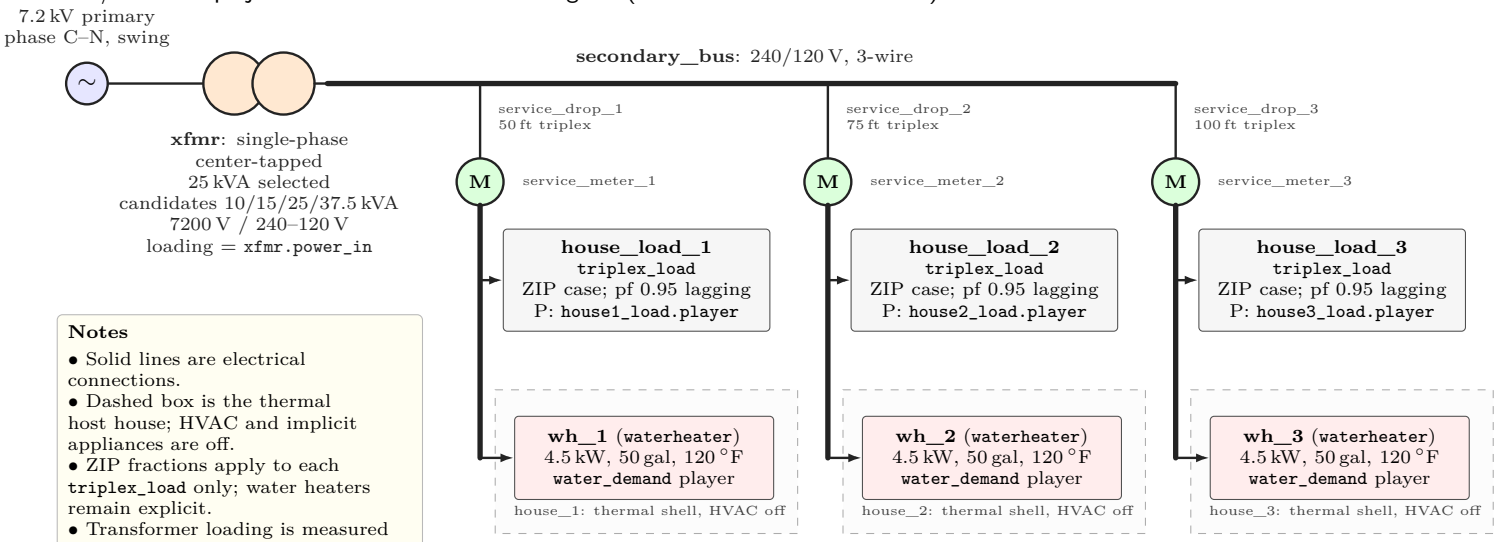
Three-House Service: Transformer, Meters, and Loads

EE 5/430 term project — GridLAB-D one-line diagram (selected transformer: 25 kVA)



Notes

- Solid lines are electrical connections.
- Dashed box is the thermal host house; HVAC and implicit appliances are off.
- ZIP fractions apply to each `triplex_load` only; water heaters remain explicit.
- Transformer loading is measured at `xfmr.power_in`.
- Service meters record combined triplex-load and water-heater power.
- ZIP cases: P, I, Z, and mixed $0.3P + 0.3I + 0.4Z$.